SIGNBRIDGE AI

AN AI-POWERED, CAMERA-BASED SIGN RECOGNITION PROTOTYPE

TEAM .....

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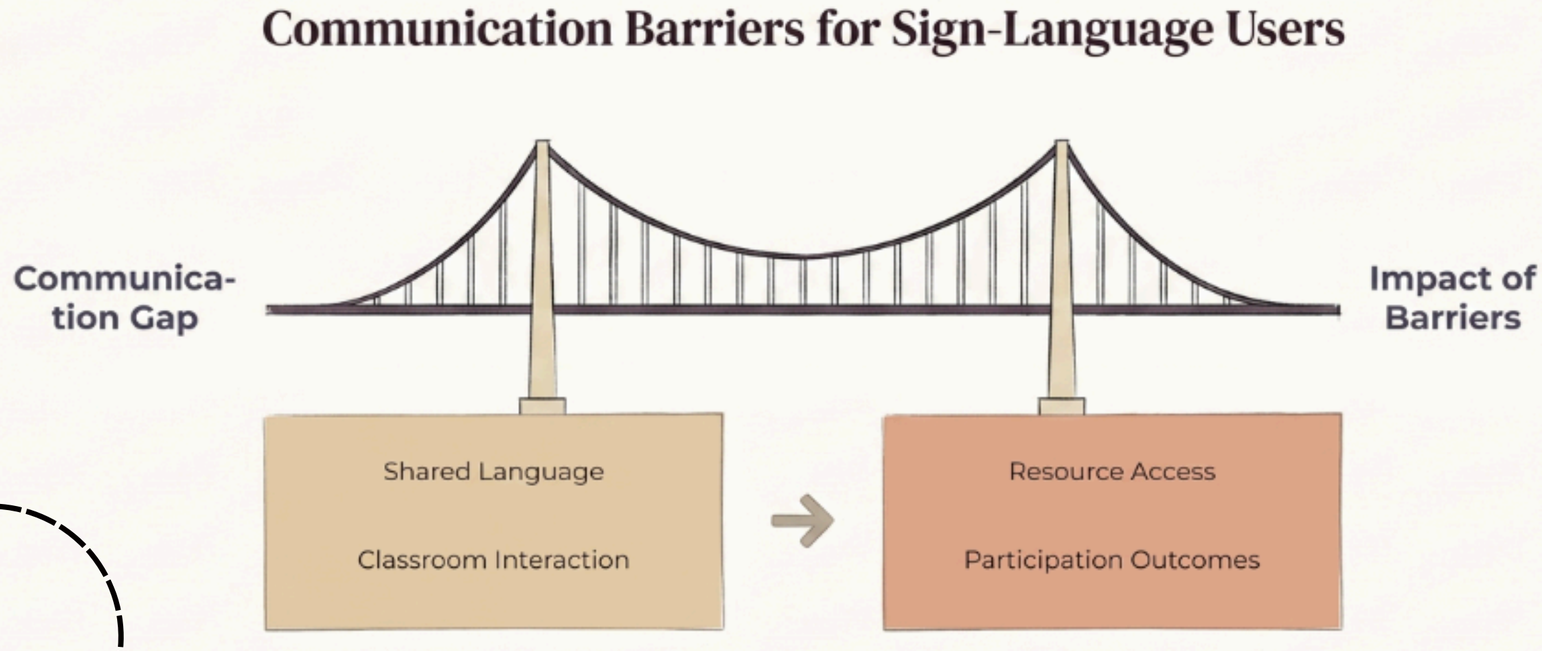
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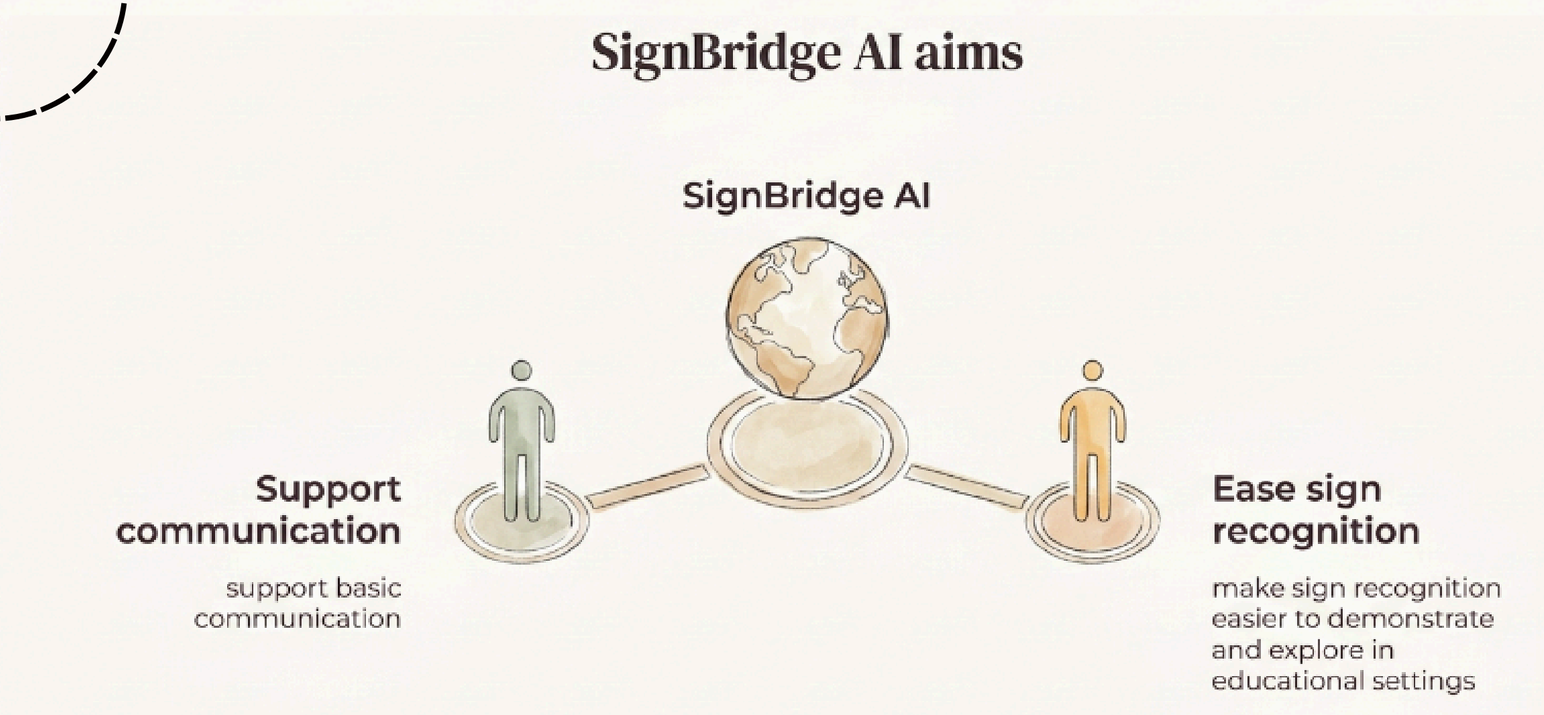


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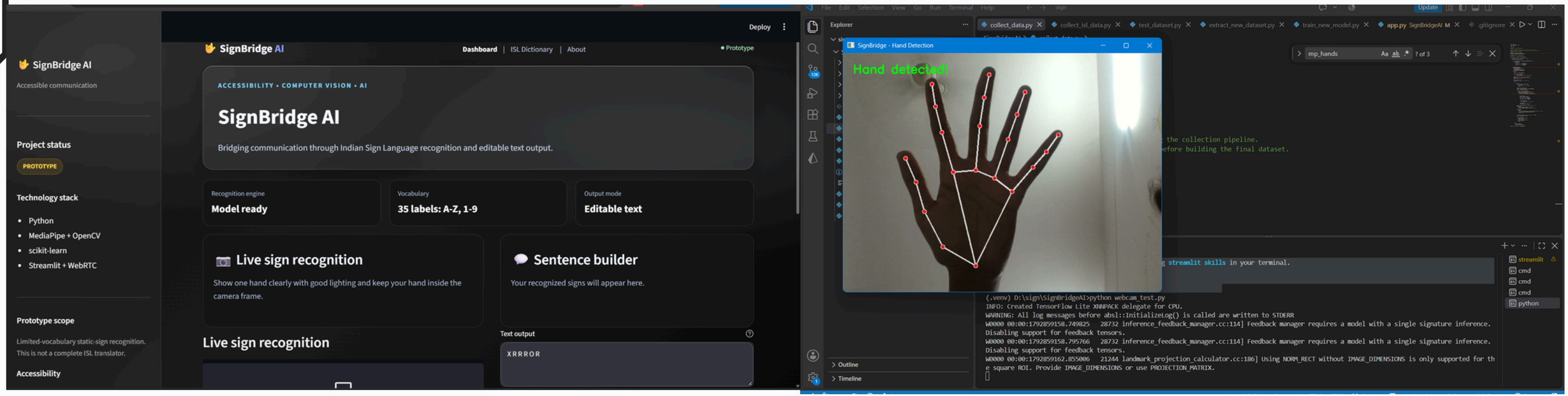
Problem statement :

How might we develop an affordable, accessible, camera-based tool that recognizes a limited set of Indian Sign Language hand signs and converts them into readable text?





INTRODUCING SIGNBRIDGE AI



Capture hand gesture

The webcam captures the user's hand gesture.

Extract hand landmarks

MediaPipe identifies hand keypoints and produces numerical features.

Predict sign label

A trained Extra Trees classifier predicts a label.

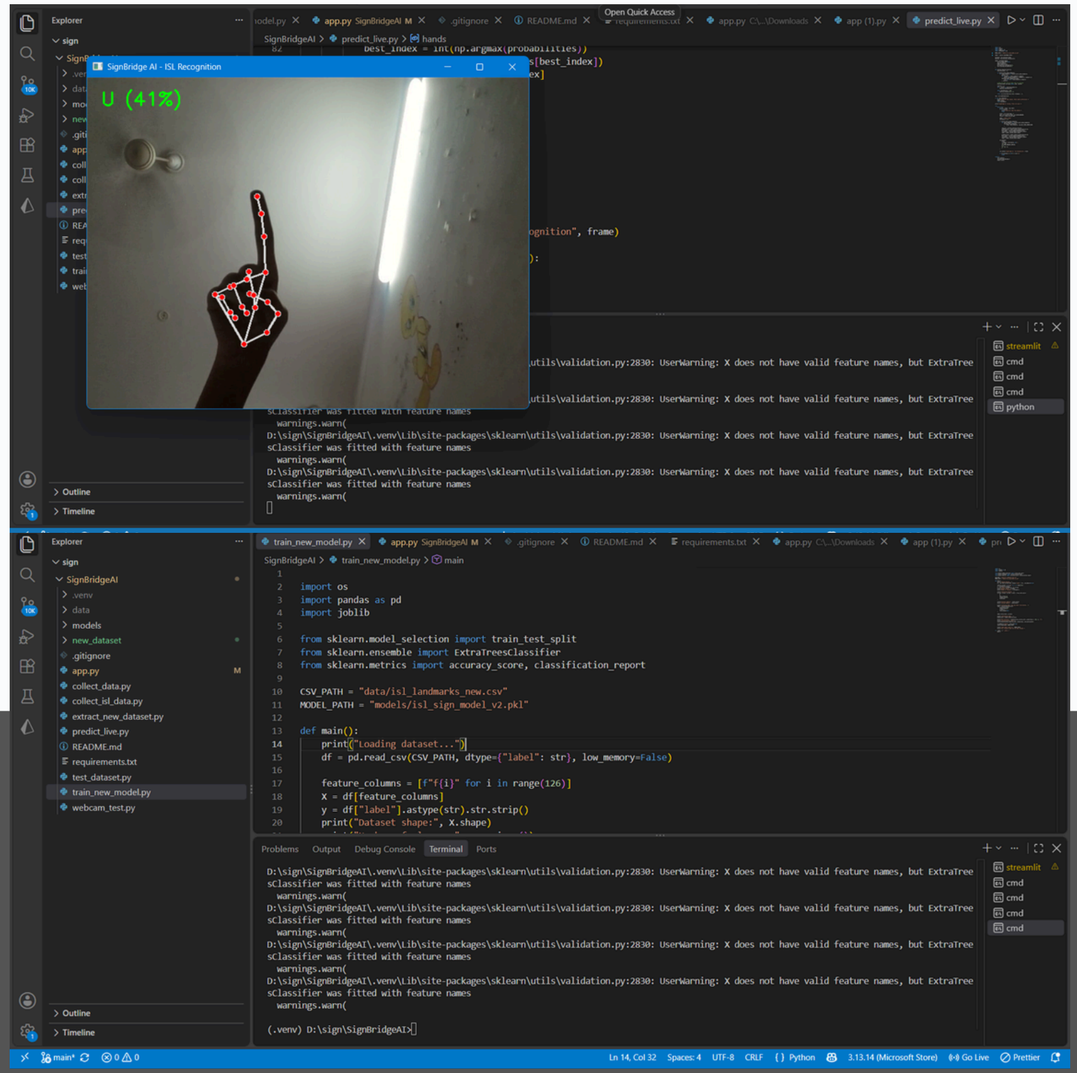
Display editable text

The dashboard displays recognized characters or digits for the user to review.

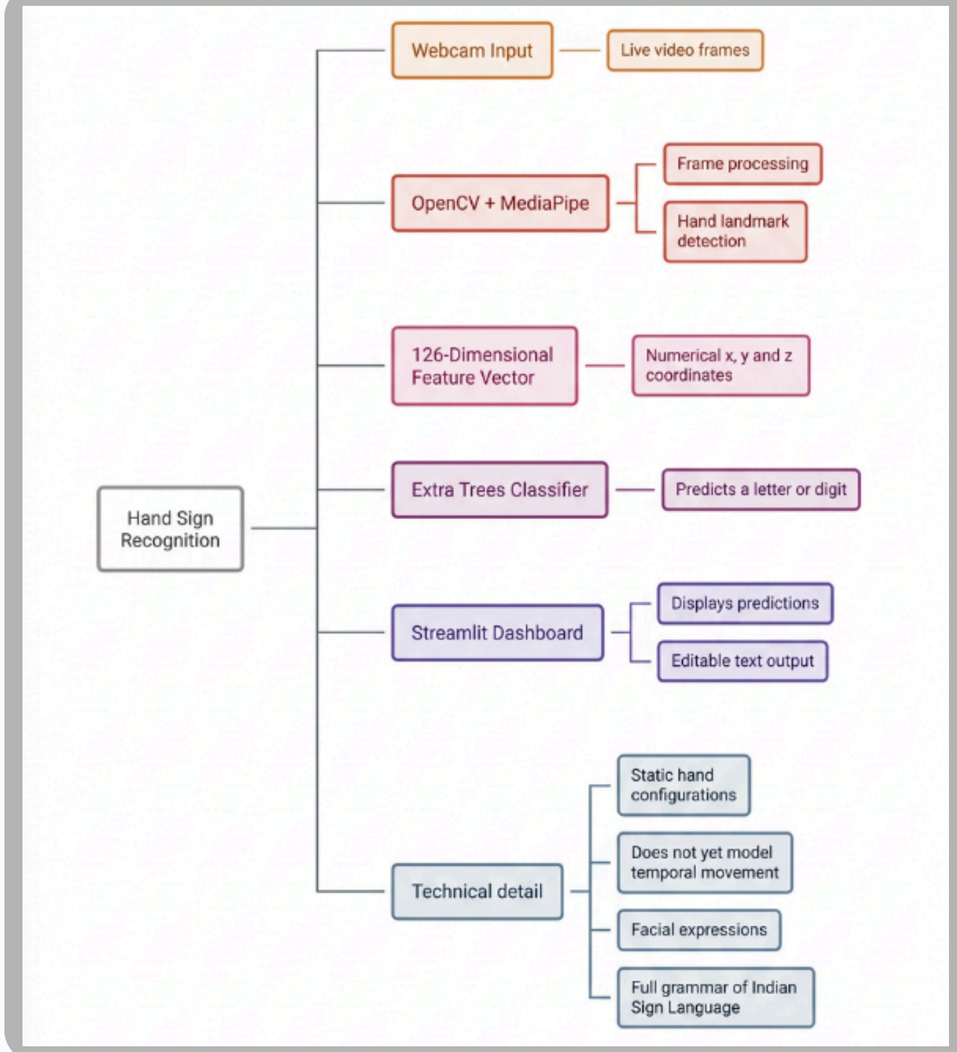
TECHNOLOGY STACK

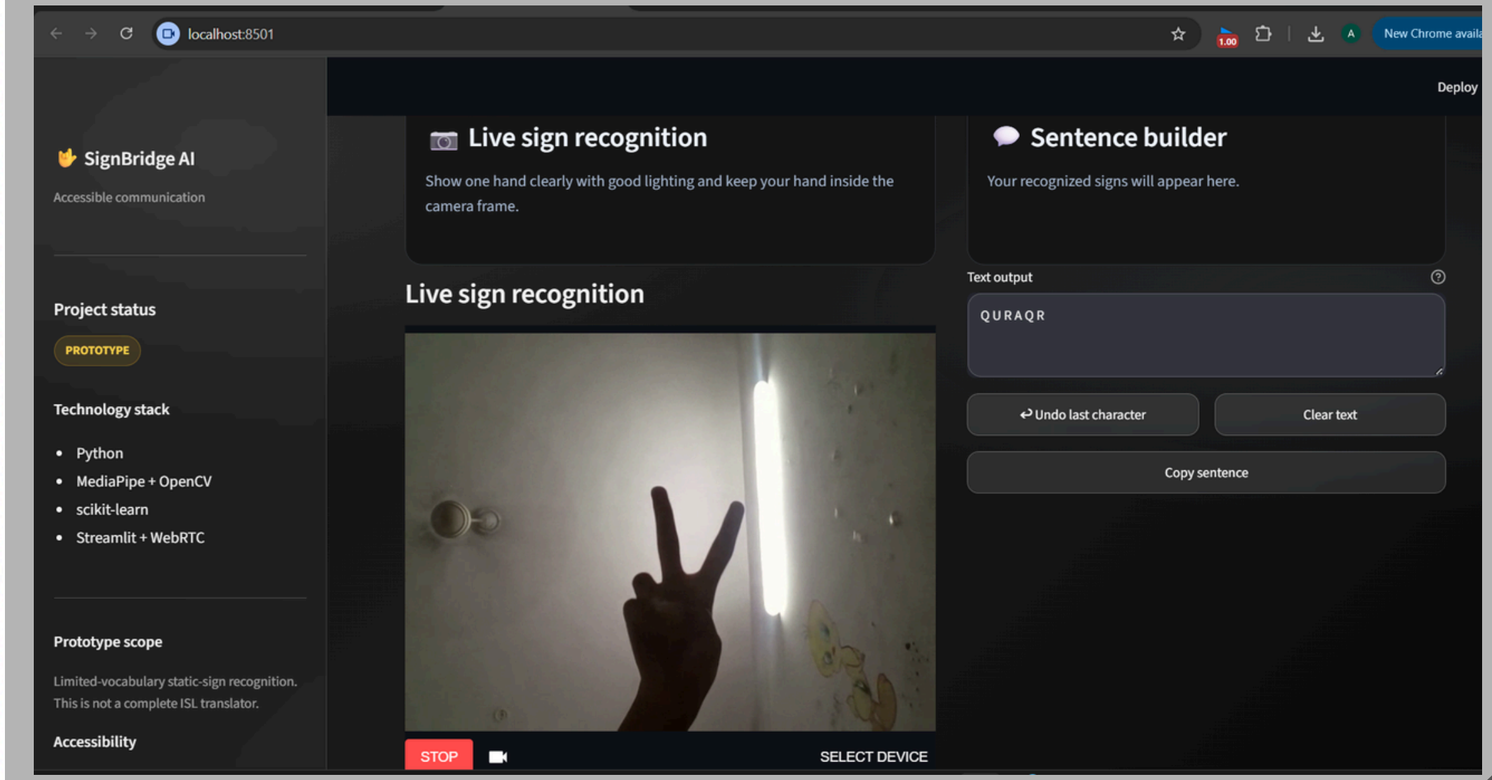
- python
- mediapipe hands
- open cv
- numpy
- pandas
- sickit learn
- streamit
- streamlit-webrtc
- joblib

Programming language	Main programming language
Detects hand landmarks	**Hand detection**
Frame processing	Processes webcam frames and images
Handles numerical landmark features	**Numerical features**
Dataset organization	Organizes the landmark dataset
Trains the Extra Trees classification model	**Model training**
Dashboard construction	Builds the interactive dashboard
Connects the webcam stream to the app	**Webcam connection**
Model management	Saves and loads the trained model



FROM HAND GESTURE TO TEXT





WHAT MAKES SIGNBRIDGE AI USEFUL?

Feature	Description
Real-time input	Uses a webcam instead of requiring users to upload an image for every prediction.
Landmark-based recognition	Uses hand keypoints as numerical features rather than feeding raw images directly into the classifier.
Interactive dashboard	Combines live camera input with a text-output area.
Editable output	Allows users to review and correct the predicted characters.
Lightweight architecture	Built using Python libraries that can run on a regular development laptop.
Extensible design	The vocabulary and recognition pipeline can be improved as more reliable data becomes available.



MAKING EVERYDAY COMMUNICATION MORE ACCESSIBLE



Education

Potential to support classroom demonstrations and basic interactions between students and people unfamiliar with signing.



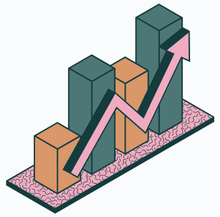
Accessibility

Explores an additional digital communication aid for selected signs.



Open-source collaboration

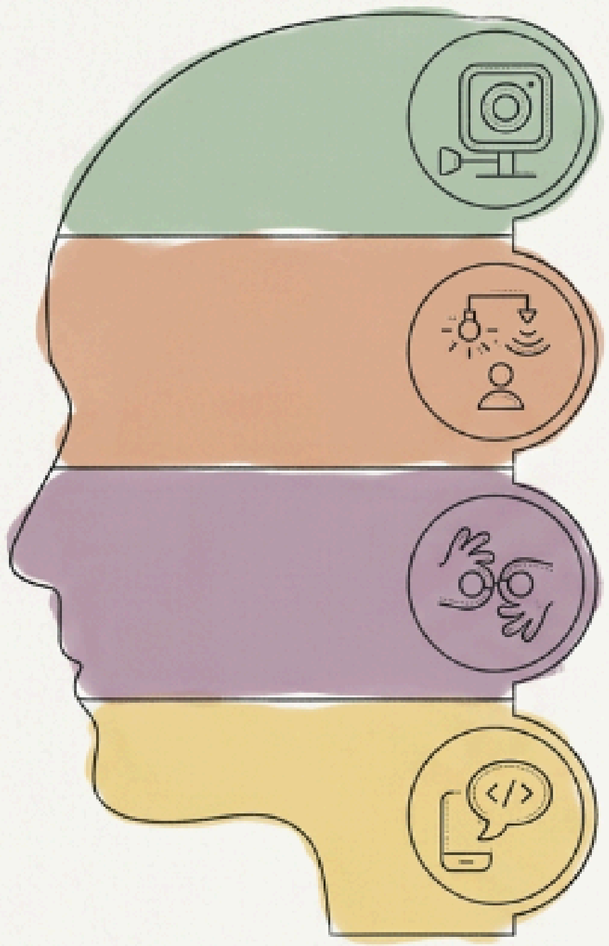
Enables contributors to improve the dataset, model, documentation, and user experience.



Scalability

Provides a foundation for expanding the vocabulary and evaluating more advanced recognition methods.

ROADMAP FOR DEVELOPMENT



Prototype Foundation

Webcam input, MediaPipe landmarks, Extra Trees classifier and editable text dashboard.

Reliability & Personalization

Test with unseen users, improve confusing signs, handle uncertain predictions and evaluate under different lighting conditions.

Beyond Alphabets

Expand the vocabulary, explore dynamic gestures and combine recognized signs into meaningful phrases.

Accessible Deployment

Explore text-to-speech, mobile-friendly access and collaboration with ISL users to validate usability.

SignBridge AI is a step towards making sign recognition more accessible through open-source technology. Our next priority is to improve prediction reliability and build a foundation for more expressive Indian Sign Language support.

THANK YOU

FOR ATTENTION



